Math Didn’t Fail You — Language Did

Presentation Outline

Last Updated: April 13, 2026
Version: 2.0 (Expanded with 18-card layered examples)

Subtitle: The barrier between “everyday people” and “advanced mathematics” is largely linguistic and psychological, not conceptual.

Tagline: A perspective on math from a girl tired of being the only one excited about it

PRESENTATION FLOW

Opening: Hook & Reality Check

THE REALITY CHECK

- Ask the room: “Who here thinks they’re bad at math?”
- Emphasize the universality of math anxiety
- Frame: This is THE INTRODUCTION to shift their perspective

Slide 1: Title Slide

Speaker Notes:

- Open with vulnerability: “I’m tired of being the only one excited about math”
- Position this as a perspective shift, not a lecture
- Promise: By the end, you’ll see you already ‘do’ calculus, abstract algebra, and topology in everyday life

“Math Didn’t Fail You — Language Did.”

- Main thesis: The barrier between “everyday people” and “advanced mathematics” is largely linguistic and psychological, not conceptual
- Set tone: personal, frustrated with status quo, evidence-based

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Slide 2: Who here thinks they’re bad at math?

Speaker Notes:

- This is the hook that engages the audience immediately
- Show of hands: ask who thinks they’re bad at math
- Promise a concrete demonstration

- Set expectation: You'll prove the thesis with a real example
- Transition: "Let's start by understanding what's really happening"

Who here thinks they're bad at math?

By the end, you'll see you already 'do' calculus, abstract algebra, and topology in everyday life. I'll prove it with one scary-looking problem that collapses to simple algebra.

THE REALITY CHECK | INTRODUCTION

Slide 3: Why Are You "Bad" at Math?

Speaker Notes:

- Don't dismiss anxiety — it's real
- But point out: the problem isn't that anxious people CAN'T do math
- The problem is they DISENGAGE
- This means: we can intervene by redesigning HOW we present math
- If we keep people engaged, anxiety's impact drops dramatically

64% report math anxiety
But anxiety alone is not the direct performance killer. The stronger mechanism is **math engagement**.

KEY QUOTE:

> "People with high math anxiety often know how to do the math — it's an emotional reaction that interferes, not a lack of skill."
> — American Psychological Association

MEDIATING MECHANISM (SPRINGER, 2025):

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Math Anxiety → Math Engagement → Math Performance
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Anxiety still matters because it suppresses engagement. If we can standardize high engagement, anxiety's performance impact drops substantially.

KEY TAKEAWAY:

> "Math anxiety" is a major indirect force, not always a direct one. Engagement is the lever we can design and control.

Supporting Points:

Direct Effect Is Weaker

- Students with anxiety do not automatically perform poorly in every condition

- Performance is more strongly tied to whether they stay cognitively engaged with the task

Engagement Mediates

- The 2025 study shows anxiety predicts lower engagement, and lower engagement predicts lower performance
- The pathway is primarily mediated through engagement

Source Citation:
"Influence of mathematics anxiety on mathematics performance: mediating effects of mathematical engagement" (Mathematics Education Research Journal, 2025). DOI: 10.1007/s13394-025-00536-1

Slide 4: The Core Problem: It's a Language Barrier

Speaker Notes:

- This is the thesis slide
- Hammer home: it's NOT about intelligence
- It's about ACCESS — specifically linguistic access
- Set up the examples coming next

Four Authoritative Quotes:

Sheila Tobias (Author of *Overcoming Math Anxiety*, 1978)

> "The barrier to math is often linguistic, not conceptual. Many students can handle the mathematical concepts but become unnerved by the unfamiliarity of the symbols, terminology, and structure."

Anna Sfard (*Thinking as Communicating*, 2008)

> "Thinking is a form of communication. Students 'hit a wall' not because they can't do the math, but because they can't parse the formal language it's written in."

American Psychological Association (*Preventing Math Anxiety*, 2023)

> "Math anxiety is an emotional problem, not a cognitive one. When anxiety is reduced, anxious individuals perform just as well as their non-anxious peers."

Ferreira, R. A., Rodriguez, C., Guzmán, B., Sepúlveda, F., & Peake, C. (*The Interplay of Vocabulary, Working Memory, and Math Anxiety*, 2025)

> "Jargon is the gatekeeper, not intelligence. The more specialized the vocabulary, the more it creates an 'insider vs. outsider' dynamic that can intimidate and alienate learners."

KEY TAKEAWAY (Dark Box - Right Side):

> Most people who say "I'm bad at math" are actually saying "math language scares me."

The ability is already there. The interface is the problem.

Slide 5: Scary Terms You Already Understand

Speaker Notes:

- This slide has TWO LAYERS of 9 cards each (18 total)
- First layer appears with fragments 1-3, then overlays with second layer (fragments 4-6)
- Move through these quickly but confidently
- Emphasize: "You ALREADY understand the concept"
- The only difference is the label
- This builds momentum and confidence

18 Examples (Two Layers):

First Layer (Cards 1-9):

1. **Taylor Series** — YOU KNOW THIS AS: **Estimated Time Arrival.** When your GPS estimates your arrival time, it is approximating future behavior from what's happening right now — recent speed and acceleration.
2. **Integral** — YOU KNOW THIS AS: **Accumulation.** Estimating the total cost of gas for a trip by adding up small chunks along the way. Summing up tiny pieces to get a total.
3. **Eigenvalue / Eigenvector** — YOU KNOW THIS AS: **Stretching a Rubber Band.** The direction along the band that only *stretches* — never rotates — is the eigenvector. How much it stretches is the eigenvalue.
4. **Combinatorics** — YOU KNOW THIS AS: **Outfit Planning.** "4 shirts and 3 pants = 12 outfits." Counting arrangements and possibilities is the heart of combinatorics.
5. **Ergodicity** — YOU KNOW THIS AS: **Shuffling Cards.** Shuffle a deck enough times and you statistically see every arrangement — the same picture as laying all possibilities out at once. You live inside ergodic processes daily.
6. **Graph Theory** — YOU KNOW THIS AS: **Social Networks.** "Friend of a friend." Finding the shortest connection path between people. Dots (people) connected by lines (relationships).
7. **Fourier Transform** — YOU KNOW THIS AS: **Hearing Music.** Your ear breaks a complex sound wave (a chord) into individual notes. That decomposition is a biological Fourier transform.
8. **Topology** — YOU KNOW THIS AS: **Shape Equivalence.** A donut and a coffee mug are "the same" because each has one hole. You intuitively group objects by their fundamental structure.
9. **Diffeomorphism** — YOU KNOW THIS AS: **Stretching Pizza Dough.** Reshaping something smoothly without tearing or gluing. A reversible transformation of shape is a diffeomorphism.

Second Layer (Cards 10-18 - Overlays):

10. **Euclidean Geometry** — YOU ALREADY KNOW: **Triangle's angles add up to 180°.** The shortest distance between two points is a straight line — these are Euclid's axioms, formalized over two thousand years ago, and you internalized them without a textbook.
11. **Abstract Algebra** — YOU ALREADY KNOW: **Symmetry everywhere.** The symmetry of a snowflake, the repeating pattern of wallpaper, the rotational symmetry of a car wheel — all are described by group theory.
12. **Epsilon-Delta Limits** — YOU ALREADY KNOW: **When the GPS counts down** — 2 miles, 1 mile, 0.5 miles, 0.1 miles — and you get to say, "We're almost there." You're watching a limit converge in real time. The destination is the limit; you approach it, getting arbitrarily close.
13. **Topoisomerases** — YOU ALREADY KNOW: **Headphone tangles.** "Spontaneous knotting" is a mathematical certainty. If a string is long enough and agitated, it will form a knot. Researchers use Jones Polynomials to study why certain cords tangle more than others.
14. **Bayesian Inference** — YOU ALREADY KNOW: **Finding your lost phone.** You search the couch where you last saw it (prior belief). Not there? A Bayesian updates the probability and decides the kitchen is now the most likely posterior location.
15. **Game Theory** — YOU ALREADY KNOW: **Four-way stop standoff.** Everyone waiting for someone else to move — you're in a "stable" state where no one gains anything by changing their strategy alone. That's high-level economics and math in a suburban intersection.
16. **Quaternions** — YOU ALREADY KNOW: **Smooth 3D rotation.** While we think in 3D, computer programs — like video games or AR filters on your phone — use 4D quaternions to calculate how objects rotate smoothly without glitching.
17. **Differential Equations** — YOU ALREADY KNOW: **Your morning coffee cooling.** Newton's Law of Cooling (a first-order differential equation) dictates how fast it hits room temperature. The hotter it is compared to the room, the faster it loses heat.
18. **Markov Chains** — YOU ALREADY KNOW: **Smartphone Text Prediction.** When your phone suggests the next word you're likely to type, it's using a Markov model. It looks at the word you just typed and calculates the most statistically likely word to follow it.

Slide 6: Research Evidence

Speaker Notes:

- This is your credibility slide
- You're not just opining — published research backs this up
- The problem is PRESENTATION, not ability
- Set up the solution section

RESEARCH EVIDENCE

The barrier isn't ability. It's how math is presented.

Two independent research reviews reach the same conclusion: symbolic format and notation density—not the underlying ideas—are the triggers for avoidance and disengagement.

Two Studies (Side by Side):

FRONTIERS IN PSYCHOLOGY — 2022
Trait anxiety is activated by format, not concept

Trait math anxiety amplifies performance failures on complex tasks—but the trigger is symbolic presentation and formal notation, not the mathematical ideas themselves.

JRSMTE LITERATURE REVIEW — 2022
Notation density directly raises cognitive load

When unfamiliar symbols don't map to a student's intuitive model, working memory fills up translating notation—leaving little capacity for actual mathematical reasoning.

Slide 7: Engagement is the Mediating Variable

Speaker Notes:

- This is the "so what?" moment
- We can INTERVENE
- If we change how we present math, we keep people engaged
- And engagement breaks the anxiety chain
- Transition: "So what do we actually DO about this?"

RESEARCH SYNTHESIS

Engagement is the mediating variable. Language controls engagement.

The strongest predictor of whether students persist in math is not ability—it's engagement quality. And engagement quality is directly shaped by language, format, and the order in which concepts are introduced. Pedagogy can break the anxiety-performance chain.

SPRINGER MERJ, 2025

SUPPORTING EVIDENCE

META-ANALYSIS (ERIC) — 2023
Language interventions are repeatable

Vocabulary support and gradual notation introduction consistently improve outcomes across different populations, settings, and math subjects.

MDPI EDUCATION SCIENCES — 2025
Vocabulary is the gatekeeper

Math-specific vocabulary is a stronger predictor of early math performance than general language skills. Students who can't parse terminology experience higher anxiety, creating a vicious cycle where jargon blocks access.

FRONTIERS IN EDUCATION — 2025
Formal notation is a barrier

Integrating symbolic notation with concrete, intuitive representations is critical. Formal notation alone significantly reduces motivation and increases intimidation—even when students understand the underlying ideas.

Slide 8: Math You Already Use Daily (Without Realizing It)

Speaker Notes:

- This slide has TWO LAYERS of 9 cards each (18 total)
- First layer appears with fragments 1-3, then overlays with second layer (fragments 4-6)
- This is the empowerment moment
- Don't just list — DEMONSTRATE
- Ask: "How many of you have done [action]?" for a few examples
- Make them realize: they're ALREADY mathematical thinkers
- The only difference is vocabulary

18 Examples (Two Layers):

First Layer (Cards 1-9):

1. **Galois Theory** — YOU DO THIS WHEN: **Dividing a pizza between friends.** If you want to cut a pizza into 8 equal slices, you are essentially solving the equation $x^8 = 1$ in the complex plane — a problem that Galois solved in the 19th century.
2. **Euclidean Geometry** — YOU DO THIS WHEN: **Parallel parking or packing a suitcase.** Judging angles, distances, and fit on flat surfaces is a core of Euclidean Geometry — every measurement you make in the physical world relies on it.
3. **Representation Theory** — YOU DO THIS WHEN: **Reading a map or a traffic sign.** Substituting a simpler stand-in that preserves the key structure of a complex situation is the core move of Representation Theory.
4. **Stochastic Process** — YOU DO THIS WHEN: **Predicting the next song or guessing a plot twist through patterns.** Any process whose next step has predictable probabilities rather than a fixed outcome is a Stochastic Process — you navigate them constantly.
5. **Linear Optimization** — YOU DO THIS WHEN: **Comparing unit prices at the grocery store.** Finding the best option within a spending constraint is Linear Optimization — the same math behind airline pricing and logistics networks.
6. **Irreducible Quintic** — YOU DO THIS WHEN: **Solving polynomial equations of degree five or higher.** The Irreducible Quintic theorem proves that complete solutions using radicals are impossible for general quintic equations — some problems have no simple formula.
7. **Chaos Theory** — YOU DO THIS WHEN: **Leaving two minutes late and having a completely different day.** Systems sensitive to tiny initial differences are the subject of

Chaos Theory — deterministic, yet practically unpredictable.

8. **Manifold** — YOU DO THIS WHEN: **Understanding the shape of the Earth.** Standing on it, the ground looks flat — but zoom out and it's a sphere. Every point on a manifold has a "locally flat" neighborhood.
9. **Differential Geometry** — YOU DO THIS WHEN: **Knowing that a flight path curves north on a flat map.** The shortest route on a sphere is a geodesic — the central object of Differential Geometry, solved by your GPS every trip.

Second Layer (Cards 10-18 - Overlays):

10. **Cardinality of the Continuum** — YOU DO THIS WHEN: **Acknowledging the sizes of infinity.** You already sense that some infinities feel bigger than others. There are infinitely many whole numbers — but also infinitely many numbers between 0 and 1 alone.
11. **Markov Chains** — YOU DO THIS WHEN: **Playing board games.** Any game determined entirely by dice is a Markov Chain. Your next position depends solely on where you are now and the roll of the dice.
12. **Genus** — YOU DO THIS WHEN: **Classifying objects based on their topological properties.** A sphere has genus 0 (no holes). A donut has genus 1. A pretzel has genus 2. When you categorize objects by their "holes," you're using the concept of genus.
13. **Ansatz** — YOU DO THIS WHEN: **Use an educated guess or assumption to simplify a problem.** Every time you estimate a tip, guess how long a drive will take, or eyeball whether furniture will fit in a room, you're using an ansatz.
14. **Heuristic** — YOU DO THIS WHEN: **Using rules of thumb or shortcuts to make decisions.** "Don't grocery shop hungry." "If it sounds too good to be true, it probably is." You use heuristics constantly — they're reliable shortcuts that don't need a formal proof.
15. **Ramsey Theory** — YOU DO THIS WHEN: **Noticing that in every large group, certain dynamics emerge.** Ramsey Theory proves that complete disorder is impossible at scale — order always appears, no matter how the pieces are arranged.
16. **Measure Theory** — YOU DO THIS WHEN: **Assigning "size" to sets.** "What's the chance of rain today?" You just assigned a measure (a probability) to a set of outcomes (weather conditions) — that's Measure Theory in action.
17. **Ergodicity** — YOU DO THIS WHEN: **Flipping a coin repeatedly.** Ergodicity is the idea that over time, the average outcome of a process will converge to its expected value. When you flip a coin many times, the proportion of heads will approach 50%.
18. **Number Theory** — YOU DO THIS WHEN: **Calculating time zones or scheduling across a 12-hour cycle.** "10am + 5 hours = 3pm" — numbers that wrap around is Number Theory's modular arithmetic in everyday form.

Slide 9: How We Fix The Interface

Speaker Notes:

- This is the ACTION slide

- These are five tangible, research-backed strategies
- Neither you nor the audience needs a PhD to implement these
- Emphasize: these aren't "nice to have" — they're PROVEN to work
- The fifth strategy (normalizing anxiety) is especially important for immediate impact

Five evidence-backed improvements any teacher can make.

Five Strategies:

1. Language First, Notation Second

Start with clear, plain English explanations before using mathematical symbols. This helps students develop a strong intuitive understanding before introducing the formal terminology.

2. Breaking Assignments and Problems into Smaller Steps

It's helpful to break down problems into smaller, manageable components. This approach promotes a sense of accomplishment with each completed part, ultimately boosting confidence in handling challenging work.

3. Define Vocabulary Explicitly

Don't assume students know mathematical jargon. Explain terms upfront to keep their focus on understanding concepts rather than getting distracted by unfamiliar words.

4. Use Familiar Anchors

Connect abstract mathematical ideas to real-life experiences. For example, use a speedometer to explain derivatives, making complex concepts more tangible and easier to grasp.

5. Acknowledge and Normalize Anxiety

Recognize that anxiety is a common barrier in learning math. Validate these feelings to show students that they don't reflect their ability to succeed.

KEY TAKEAWAY:

> "It is time to stop gatekeeping intelligence behind jargon. Mathematical thinking is not a rare talent — it is a universal human ability."

Slide 10: You're Not Bad at Math

Speaker Notes:

- This is the emotional landing
- Full circle back to the opening
- Reframe: it's not about ability, it's about ACCESS
- Leave them empowered, not lectured

- End on the call to action: change how you talk about math, to yourself and others

You're not bad at math.

You already think mathematically every day. Learn the dialect, and the field opens up.

STOP SAYING
~~"I'm bad at math."~~

↓

START SAYING
"I'm learning the language."

THANK YOU ●

RESEARCH SOURCES

Statistics on Math Anxiety

- 64% of Americans experience math anxiety (13% severe)
- Women: 70% vs. Men: 57%
- 47% say it held them back academically/professionally
- 37% say it hinders financial decision-making

Sources:

- Prodigy Game
- District Administration

Key Academic Sources

Sheila Tobias - *Overcoming Math Anxiety* (1978)

- Foundational work establishing math anxiety as a linguistic barrier

Anna Sfard - *Thinking as Communicating* (2008)

- Mathematics as discourse/language

APA - *Preventing Math Anxiety* (2023)

- Math anxiety is emotional, not cognitive

Ferreira et al. MDPI Education Sciences - *The Interplay of Vocabulary, Working Memory, and Math Anxiety* (2025)

- Math-specific vocabulary as gatekeeper
- Math-specific vocabulary as stronger predictor than general language skills
- Jargon creates vicious cycle of anxiety and blocked access

Frontiers in Psychology - *Unraveling the Role of Math Anxiety* (2022)

- State vs. trait anxiety; presentation as trigger

Springer MERJ - *Influence of Mathematics Anxiety on Performance* (2025)

- Engagement as mediating variable
- Language-first pedagogy buffers anxiety effects

ERIC Meta-Analysis (2023)

- Language interventions consistently effective

Frontiers in Education (2025)

- Integration of symbolic notation with concrete representations is critical
- Formal notation alone reduces motivation and increases intimidation

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APPENDIX: Extended Examples (For Q&A or Longer Format)

Additional "Scary Terms You Already Understand"

Note: Many previously "extended" examples have been promoted to the main presentation in Slides 5 and 8. The following remain as backup content for Q&A or extended formats.

Asymptote

- A line you approach but never reach
- "Almost there but never quite" — like halving distance to a wall forever

Universal Quantifier (∀)

- Symbol meaning "for all" / "every"
- "Every restaurant here is expensive" — you use this in speech constantly

Orthonormal

- Perpendicular directions at right angles, evenly scaled
- Every map grid, every 3D graph axis since middle school

Ring Theory

- Algebraic structures with addition and multiplication
- Every arithmetic operation on integers is ring theory

Module Theory

- Vector spaces where scalars come from rings
- Budget spreadsheet with different multipliers per category

CLOSING THOUGHT

The goal of this presentation is not to teach math — it's to **change the conversation about who can do math**.

When we shift from "I'm bad at math" to "I'm learning the language," we remove the stigma and open the door.

Math didn't fail you. Language did.

And language can be learned.